

Chapter 1 Coalgebras

1.1 (Associative) Algebras, and Coalgebras

1.1.1 Review: Algebras

Definition 1.1.1.1 An (associative) algebra is a triple (A, m, η) with:

- A is k -vector space;
- Operation $m : A \otimes_k A \rightarrow A$; Bilinear.
- Operation $\eta : k \rightarrow A$ which is linear.

And we have

$$\begin{aligned} A \otimes A \otimes A &\xrightarrow{\cong} (A \otimes A) \otimes A \xrightarrow{m \otimes id_A} A \otimes A \xrightarrow{m} A \\ &= \\ A \otimes A \otimes A &\xrightarrow{\cong} A \otimes (A \otimes A) \xrightarrow{id_A \otimes m} A \otimes A \xrightarrow{m} A \end{aligned}$$

And the composition

$$\begin{aligned} A &\cong k \otimes A \xrightarrow{\eta \otimes id_A} A \otimes A \xrightarrow{m} A \\ A &\cong A \otimes k \xrightarrow{id_A \otimes \eta} A \otimes A \xrightarrow{m} A \end{aligned}$$

are identity. i.e.

$$\begin{aligned} m \circ (m \otimes id_A) &= m \circ (id_A \otimes m) \\ m \circ (\eta \otimes id_A) &= m \circ (id_A \otimes \eta) \end{aligned}$$

1.1.2 Opposite Algebras

Definition 1.1.2.1 Let (A, m, η) be an algebra over k . The algebra

$$(A, m^{op}, \eta)$$

is the **opposite algebra**.

By definition, the product m^{op} is defined as $a \bullet b := b \cdot a$.

Thus

Definition 1.1.2.2 An algebra is **commutative** iff $m^{op} = m$.

1.1.3 Coalgebras

Definition 1.1.3.1 Take the following notions: Given algebra (A, m, η) , then the coalgebras (C, Δ, ϵ) ,

$$C \xrightarrow{\Delta} C \otimes C \xrightarrow{\Delta \otimes id_C} (C \otimes C) \otimes C \simeq C \otimes C \otimes C$$

$$C \xrightarrow{\Delta} C \otimes C \xrightarrow{id_C \otimes \Delta} C \otimes (C \otimes C) \simeq C \otimes C \otimes C$$

And requiring

$$C \xrightarrow{\Delta} C \otimes C \xrightarrow{\epsilon \otimes id_C} k \otimes C \simeq C$$

$$C \xrightarrow{\Delta} C \otimes C \xrightarrow{Id_C \otimes \epsilon} C \otimes k \simeq C$$

be the identity map I_C . i.e.

$$(\Delta \otimes Id_C) \circ \Delta = (Id_C \otimes \Delta) \circ \Delta$$

$$(\epsilon \otimes I_C) \circ \Delta = I_C = (I_C \otimes \epsilon) \circ \Delta$$

Definition 1.1.3.2 (Opposite Coalgebra)

Definition 1.1.3.3 (Commutative Coalgebra)

Sweedler's Notation

Definition 1.1.3.4 Let $c \in C$, then $\Delta(c) \in C \times C$ could be denoted as a symbolic sum: Suppose $\Delta(c) = \sum_{i=1}^n a_i \otimes b_i$

$$\Delta(c) := \sum_{(c)} c_{(1)} \otimes c_{(2)} \quad (\text{Heymann-Sweedler})$$

Then, the action on the first item: (For example)

$$() \Delta(c)$$

Subspace of Coalgebras

Simple Coalgebras

Grouplinke Coalgebra

Definition 1.1.3.5 ...

Coalgebra-Map

Coideals of a Coalgebra, Quotient of Coalgebra

Definition 1.1.3.6 ...

The Fundamental Theorem of Coalgebras

1.1.4 Comatrix Coalgebra

1.2 The Dual Algebras/Coalgebras

1.3 Double Duals

1.4 Wedge Product

1.5 Cofree Coalgebra on a Vector Space

Chapter 2 Representation of Coalgebras

2.1 Rational Modules of the Dual Algebra

Chapter 3 Co-Radical Filtration and Related Structures

Chapter 4 Bialgebras

Definition 4.0.0.1 ...

Chapter 5 The Convolution Algebras

◇ **Lemma 5.0.0.1** Let A be a vector space over k . With:

- Algebra structure (A, m, η) ;
- Coalgebra structure (A, Δ, ϵ) .

Give $A \otimes A$ the tensor product algebra, and coalgebra structures, then the following are equivalent:

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◦ **Proof ...**

□

Definition 5.0.0.2 Define: The **bialgebra** over the field, or a k -**bialgebra**, is a tuple

$$(A, m, \eta, \Delta, \epsilon)$$

where:

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Chapter 6 Hopf Algebras

6.1 Definition

Definition 6.1.0.1 Define:

♣ **Example 6.1.0.2** Group Algebra $K[G]$: Suppose G be a group. Then, consider $K[G]$ with following structures:

- $m : (g, h) \mapsto gh$;
- $u : K \ni 1 \mapsto 1_G$;
- $\Delta : g \mapsto g \otimes g$; (the)
- $\epsilon(g) := 1_K$; (Counit)
- $S : g \mapsto g^{-1}$; (Antipode)